Advanced Topics in Computer Graphics I

BRDF Models

Prof. Dr. Reinhard Klein

Institute of Computer Science II - Computer Graphics
University of Bonn

May 13, 2026

The Rendering Equation is defined as

$$\boldsymbol{L}_o(\boldsymbol{x}, \boldsymbol{\Omega}_o) = \boldsymbol{L}_e(\boldsymbol{x}, \boldsymbol{\Omega}_o) + \int_{\mathcal{H}_i} \boldsymbol{f}_{BSDF}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o)\, \boldsymbol{L}_i(\boldsymbol{x}, \boldsymbol{\Omega}_i) \cos(\theta_i)\, d\omega_i$$

where

$$\boldsymbol{f}_{BSDF}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o) = \frac{\boldsymbol{w}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o)}{\cos(\theta_o)} \quad \left[\frac{1}{\mathsf{sr}}\right]$$

is called the Bidirectional Scattering Distribution Function and defines the amount of light that is reflected/refracted in direction $\boldsymbol{\Omega}_o$ coming from $\boldsymbol{\Omega}_i$.

Now: For the sake of simplicity, we now assume that the considered materials only reflect (true for most materials), so the BSDF reduces to a BRDF!

So first, we need some properties for BRDF's:

- Ray Tracers normally trace light paths back to the light source. What does this mean for the BRDF?
- Intuitively, it is not possible that more light gets reflected than hitting the surface. What is a physically plausible BRDF?

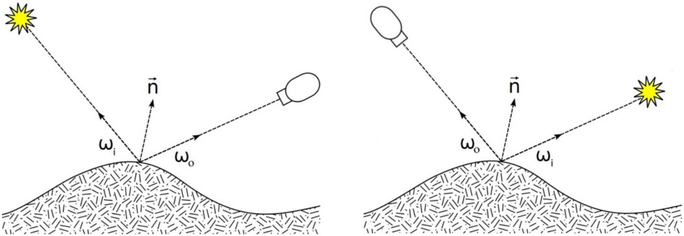
Ray Tracers normally trace light paths back to the light source. What does this mean for the BRDF?

Definition (Helmholtz Reciprocity[1])

A BRDF $\boldsymbol{f}_{BRDF}$ is called *reciprocal* if

$$\boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o) = \boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_o, \boldsymbol{x}, \boldsymbol{\Omega}_i)$$

for all $\boldsymbol{\Omega}_i, \boldsymbol{\Omega}_o \in \mathcal{H}$ and $\boldsymbol{x} \in \mathcal{S}$.



[1] M. Minnaert. "The reciprocity principle in lunar photometry". In: *The Astrophysical Journal* 93 (1941), pp. 403–410.

Intuitively, it is not possible that more light gets reflected than hitting the surface. What is a physically plausible BRDF?

Definition (Albedo)

The *Total Reflectance* or *Albedo* is defined as

$$\begin{aligned}\boldsymbol{a}(\boldsymbol{x}, \boldsymbol{\Omega}_i) &= \mathrm{Pr}(\text{photon gets reflected} \mid \text{photon is coming from } \boldsymbol{\Omega}_i) \\ &= \int_{\mathcal{H}} \boldsymbol{w}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o)\, d\omega_o = \int_{\mathcal{H}} \boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o) \cos(\theta_o)\, d\omega_o\end{aligned}$$

Definition (Physically Plausible[2])

A BRDF $\boldsymbol{f}_{BRDF}$ is called *physically plausible* if

$$\boldsymbol{a}(\boldsymbol{x}, \boldsymbol{\Omega}_i) = \int_{\mathcal{H}} \boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_i, \boldsymbol{x}, \boldsymbol{\Omega}_o) \cos(\theta_o)\, d\omega_o \leq \mathbf{1}$$

for all $\boldsymbol{\Omega}_i \in \mathcal{H}$ and $\boldsymbol{x} \in \mathcal{S}$.

[2] R. R. Lewis. "Making shaders more physically plausible". In: *Computer Graphics Forum*. Vol. 13. 2. 1994, pp. 109–120.

If the BRDF is reciprocal, then the albedo can also be interpreted as the optical response of the surface to homogeneous sky-light illumination of unit intensity:

$$\begin{aligned}(\boldsymbol{T}\mathbf{1})(\boldsymbol{x},\boldsymbol{\Omega}_o) &= \int_{\mathcal{H}} \boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_i,\boldsymbol{x},\boldsymbol{\Omega}_o)\cdot\mathbf{1}\cdot\cos(\theta_i)\,d\omega_i \\ &= \int_{\mathcal{H}} \boldsymbol{f}_{BRDF}(\boldsymbol{\Omega}_i,\boldsymbol{x},\boldsymbol{\Omega}_o)\cos(\theta_o)\,d\omega_o \\ &= \boldsymbol{a}(\boldsymbol{x},\boldsymbol{\Omega}_i)\end{aligned}$$

Since we now know important properties of BRDF's, how to get one?

- **Measurement data**: Tables (4D) or coefficients for a set of (basis) functions
- **Phenomenological models**: Qualitative approach with intuitive parameters
- **Simulation**: Numerical simulation of reflection properties by modeling known micro geometry (Maxwell equations, micro facet models)

Note: All BRDF's – which we will consider in the following – do not directly depend on the position $\boldsymbol{x}$, so they are 4-dimensional!

Definition (Isotropic BRDF)

A BRDF $\boldsymbol{f}_{BRDF}$ is called *isotropic* if it is invariant under rotations R_ϕ of both the incoming and outgoing direction around the surface normal $\boldsymbol{n}$:

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{f}_{BRDF}(R_\phi\, \boldsymbol{l},\, R_\phi\, \boldsymbol{v}) \qquad \forall\, \phi.$$

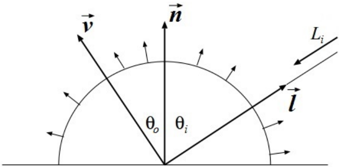
Otherwise the BRDF is called *anisotropic*.

Consequence for the parameterisation. A general BRDF depends on four angular parameters $(\theta_i, \phi_i, \theta_o, \phi_o)$. For an isotropic BRDF only the *relative* azimuth $\Delta\phi = \phi_i - \phi_o$ matters, reducing the domain to three dimensions $(\theta_i, \theta_o, \Delta\phi)$.

Half-vector form. For microfacet models this further simplifies: because D and G depend only on θ_h (not ϕ_h) in the isotropic case, the effective domain collapses to (θ_h, θ_d), where θ_d is the angle between $\boldsymbol{l}$ and $\boldsymbol{h}$.

Examples. Brushed metal, hair, and satin are anisotropic. Most paints, plastics, and skin are well approximated as isotropic.

First of all, consider diffuse – optically very rough – surfaces reflecting a portion of the incoming light with radiance uniformly distributed in all directions.



Looking at the wall, sand, etc. the perception is the same regardless of the viewing direction.

If the BRDF is independent of the viewing direction $\boldsymbol{v}$, it must also be independent of the light direction $\boldsymbol{l}$ because of the Helmholtz Reciprocity. Thus, the BRDF of these diffuse surfaces is constant:

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{c}$$

Note: From now on, we use $\boldsymbol{l}$ for incoming and $\boldsymbol{v}$ for outgoing direction! All vectors point away from the surface!

Thus, the BRDF of these diffuse surfaces is constant:

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{c}$$

The albedo of a diffuse reflection is

$$\begin{aligned}\boldsymbol{a}(\boldsymbol{l}) &= \int_{\mathcal{H}} \boldsymbol{c} \cdot \cos(\theta_o)\, d\omega_o \\ &= \boldsymbol{c} \cdot \int_{\phi_o=0}^{2\pi} \int_{\theta_o=0}^{\pi/2} \cos(\theta_o) \sin(\theta_o)\, d\theta_o\, d\phi_o \\ &= \boldsymbol{c} \cdot \pi\end{aligned}$$

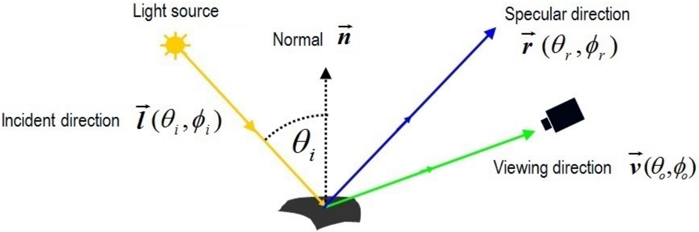
Definition (Diffuse BRDF[3])

The *Diffuse BRDF* or *Lambert-BRDF* is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \frac{\boldsymbol{\kappa}_d}{\pi} \quad , \quad \boldsymbol{\kappa}_d \leq \mathbf{1}$$

[3] J. Lambert and E. Anding. *Photometria, sive De mensura et gradibus luminis, colorum et umbrae (1760)*. Ostwalds Klassiker der exakten Wissenschaften. W. Engelmann, 1892.

An ideal mirror complies with the reflection law of geometric optics, which states that incoming direction $\boldsymbol{l}$, surface normal $\boldsymbol{n}$ and outgoing direction $\boldsymbol{r}$ are in the same plane, and incoming angle θ_i equals to outgoing angle θ_r.



An ideal mirror reflects only into the ideal reflection direction $\boldsymbol{r}$. The BRDF can thus be defined by a Dirac-delta function

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{c} \cdot \frac{\delta(1 - \langle\, \boldsymbol{r} \,|\, \boldsymbol{v}\, \rangle)}{\cos(\theta_i)}$$

So light is exclusively reflected towards the viewing direction $\boldsymbol{v}$ if and only if $\boldsymbol{v} = \boldsymbol{r}$.

To make the BRDF physically plausible, we have to compute the constant $\boldsymbol{c}$:

$$\boldsymbol{a}(\boldsymbol{l}) = \int_{\mathcal{H}} \boldsymbol{c} \cdot \frac{\delta(1 - \langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle)}{\cos(\theta_i)} \cdot \cos(\theta_o)\, d\omega_o = \boldsymbol{c}$$

since the only contribution in the integral is the case where $\boldsymbol{v} = \boldsymbol{r}$ and therefore $\cos(\theta_o) = \cos(\theta_r) = \cos(\theta_i)$. This leads to the final BRDF

Definition (Ideal Mirror-like BRDF)

The *Ideal Mirror-like BRDF* is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot \frac{\delta(1 - \langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle)}{\cos(\theta_i)} \quad , \quad \boldsymbol{\kappa}_s \leq \mathbf{1}$$

Similarly we define the BSTF at a perfectly smooth surface.

Definition (Ideal BTDF)

The *Ideal BTDF* is defined as

$$\boldsymbol{f}_{BTDF}(\boldsymbol{l},\boldsymbol{v}) = \boldsymbol{\kappa}_t \cdot \frac{\delta(1 - \langle\, \boldsymbol{t} \,|\, \boldsymbol{v} \,\rangle)}{\cos(\theta_i)} \quad , \quad \boldsymbol{\kappa}_t \leq \mathbf{1},$$

where the ideal transmission vector $\boldsymbol{t}$ is computed from the incoming direction $\boldsymbol{l}$ using Snell's law, i.e. that $\boldsymbol{l}, \boldsymbol{n}, \boldsymbol{t}$ are colinear and that $n_i \sin(\theta_i) = n_t \sin(\theta_t)$. ($n_i, n_t$ are the refractions indices of the media at the side of the incoming and the side of the refracted light.)
Given the absorption coefficient, the index of refraction and the angle which the incident ray makes with the surface normal, the Fresnel equations specify the material's corresponding reflectance $\boldsymbol{\kappa}_s$ and transmittance $\boldsymbol{\kappa}_t$ for two different polarization states of the incident illumination.

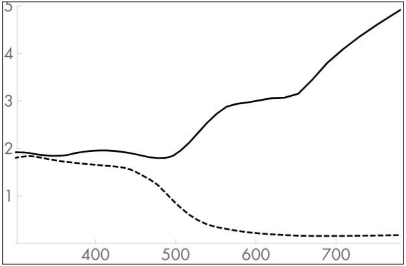


Figure: Absorption coefficient and index of refraction of gold. This plot shows the spectrally varying values of the absorption coefficient k (solid line) and the index of refraction (dashed line) n for gold, where the horizontal axis is the wavelength in nm.

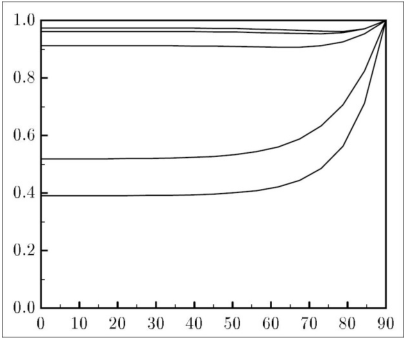


Figure: Reflectivity of gold for wavelengths 400, 500, 600, 700, and 800 nm. (bottom to top). Curves were generated directly from Fresnel Equations.

Images from[4].

[4] P. S. Shirley. "PHYSICALLY BASED LIGHTING CALCULATIONS FOR COMPUTER GRAPHICS". en. In: (), p. 183.

We destinguish between important classes of materials:

- **Conductors**, or metals are opaque and reflects back a significant portion of the illumination. A portion of the light is also transmitted into the interior of the conductor, where it is rapidly absorbed: total absorption typically occurs within the top $0.1\mu m$ of the material. Only extremely thin metal films are capable of transmitting appreciable amounts of light. They are described by their refractive index n and their extinction coefficient k.
- **Dielectrics** are materials that don't conduct electricity. They transmit a portion of the incident illumination. Examples are glass, mineral oil, water, and air.

A dielectric is characterized by its index of refraction n. It measures how much an electromagnetic wave slows down relative to its speed in vacuum. The extinction coefficient of dielectrics is (by definition) zero.

Medium	Index of refraction n
Vacuum	1.0
Air at sea level	1.00029
Water ($20\,^\circ$C)	1.31
Glass	1.5-1.6
Diamond	2.4

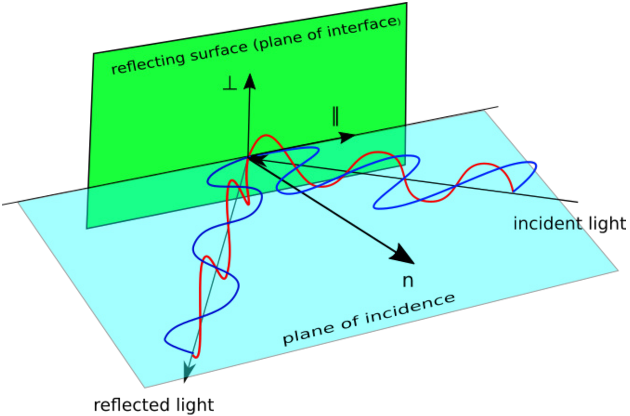


Figure: Definition of $\perp$ and $\|$ direction at an interface.

The Fresnel reflectance at the boundary to a dielectric-conductor interface is given by[5]

$$F_{\perp}(\lambda,\theta) = \frac{a^2 + b^2 - 2a\cos(\theta) + \cos(\theta)^2}{a^2 + b^2 + 2a\cos(\theta) + \cos(\theta)^2}$$

$$F_{\parallel}(\lambda,\theta_i) = \frac{a^2 + b^2 - 2a\sin(\theta)\tan(\theta) + \sin(\theta)^2\tan(\theta)^2}{a^2 + b^2 + 2a\sin(\theta)\tan(\theta) + \sin(\theta)^2\tan(\theta)^2}$$

where $F_{\perp}$ und $F_{\parallel}$ are the reflectivities for the two planes of polarization, θ is the angle of incidence and a and b are given by

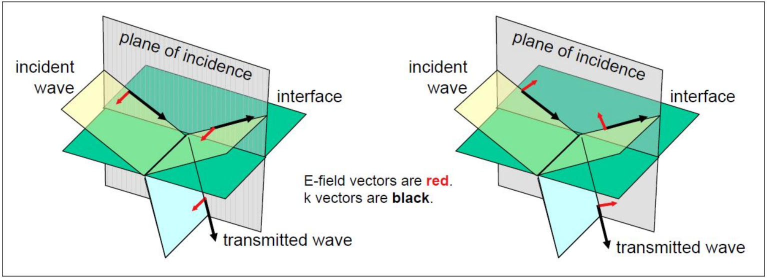
$$a^2 = \frac{1}{2n_i^2}\left(\sqrt{\left(n^2 - k^2 - n_i^2\sin((\theta)^2\right)^2 + 4n^2k^2} + n^2 - k^2 - n_i^2\sin(\theta)^2\right)$$

$$b^2 = \frac{1}{2n_i^2}\left(\sqrt{\left(n^2 - k^2 - n_i^2\sin((\theta)^2\right)^2 + 4n^2k^2} - n^2 + k^2 + n_i^2\sin(\theta)^2\right)$$

where n_i is the refraction index of the dielectric and n and k the refractive index and extinciton coefficient of the conductor, respectively.

[5] P. S. Shirley. "PHYSICALLY BASED LIGHTING CALCULATIONS FOR COMPUTER GRAPHICS". en. In: (), p. 183.

The Fresnel equations are usually only written for the interface between two dielectrics. In this case the extinction coefficient of the second dielectric is $k = 0$ and it's index of refraction is n_t:



$\perp$**-polarized** $\quad$ $\|$**-polarized**

$$F_{\perp}(\lambda, \theta_i) = \frac{n_i \cos(\theta_i) - n_t \cos(\theta_t)}{n_i \cos(\theta_i) + n_t \cos(\theta_t)} \qquad F_{\|}(\lambda, \theta_i) = \frac{n_t \cos(\theta_i) - n_i \cos(\theta_t)}{n_t \cos(\theta_i) + n_i \cos(\theta_t)}$$

and, for both polarizations $n_i \sin(\theta_i) = n_t \sin(\theta_t)$.

A common assumption in Computer Graphics is that light is unpolarized; that is, it is randomly oriented with respect to the light wave.
With this simplifying assumption, the **Fresnel reflectance** is the average of the squares of the parallel and perpendicular polarization terms:

$$\boldsymbol{\kappa}_s = F(\lambda, \theta_i) = \frac{F_\perp(\lambda, \theta_i)^2 + F_\parallel(\lambda, \theta_i)^2}{2}$$

The transmission coefficients are given by

$\perp$-polarized

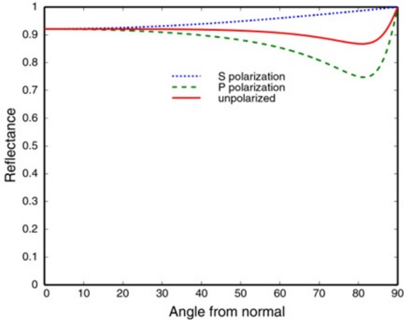
$$T_\perp(\lambda, \theta_i) = \frac{2n_i \cos(\theta_i)}{n_i \cos(\theta_i) + n_t \cos(\theta_t)}$$

$\parallel$-polarized

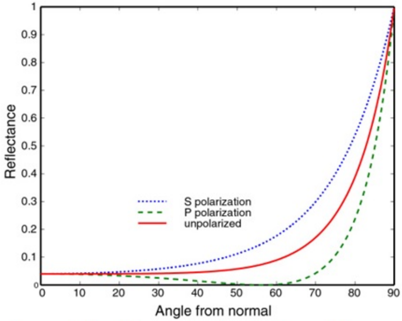
$$T_\parallel(\lambda, \theta_i) = \frac{2n_i \cos(\theta_i)}{n_t \cos(\theta_i) + n_i \cos(\theta_t)}$$

For the **transmittance** we get

$$\boldsymbol{\kappa}_t = T(\lambda, \theta_i) = \frac{T_\perp(\lambda, \theta_i)^2 + T_\parallel(\lambda, \theta_i)^2}{2}$$



Example conductor: interface vakuum -Al

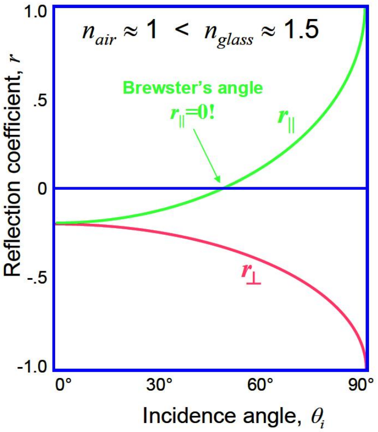


Example dielectric material: interface vakuum-glass

- The two polarizations are indistiguishable at $\theta = 0$
- Total reflection at $\theta = 90^\circ$ for both polarizations.
- Zero reflection for parallel polarization at **Brewster's angle**. It depends on the value of the ratio n_i/n_t:

$$\theta_{Brewster} = \arctan(n_i/n_t)$$

For air to glass ($n_{glas} = 1.5$) this is 56.3°.



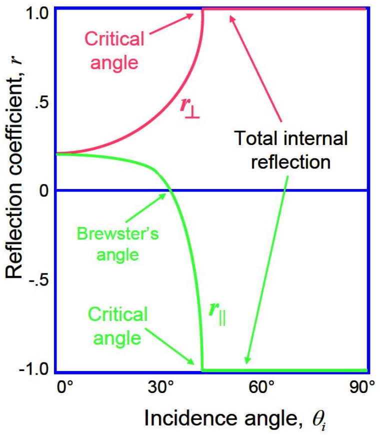
- Due to

$$n_{glas} > n_{air}$$

  we get total internal reflection above the **critical angle** θ_{crit}
- For glass-to-air

$$\theta_{crit} = \arcsin n_t/n_i \approx 41.8^\circ$$

- Note, that the sine in Snell's Law can't be greater than one!



When $\theta_i = 0$ the Fresnel equations reduce to

$$\boldsymbol{\kappa_s} = F = \left(\frac{n_t - n_i}{n_t + n_i}\right)^2$$

$$\boldsymbol{\kappa_t} = T = \frac{4 n_t n_i}{(n_t + n_i)^2}$$

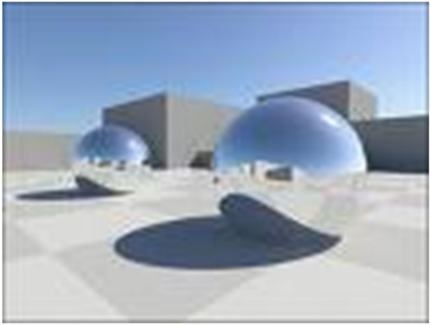
For an air-glass interface ($n_i = 1$ and $n_t = 1.5$),

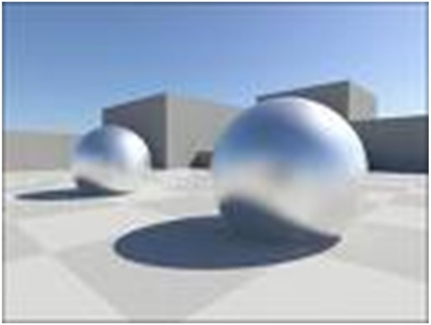
$F = 4\%$ and $T = 96\%$

- The values are the same, whichever direction the light travels, from air to glass or from glass to air.
- This 4% value has big implications for photography since it generates **lens flare effects.**

Observation: Most materials reflect specular, but not ideally specular.

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot \frac{\delta(1 - \langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle)}{\cos(\theta_i)}$$





Idea: Replace the Dirac-delta function by some other "wider" function.

Idea: Replace the Dirac-delta function by some other "wider" function.

The Dirac-delta function can be seen as the limit of a normal distribution with mean $\mu = 0$ and variance σ^2:

$$\delta(x) = \lim_{\sigma^2 \to 0} \mathcal{N}(x \mid 0\,, \sigma^2)$$

Thus use a normal distribution instead of the Dirac-delta:

Definition (Gauss-BRDF)

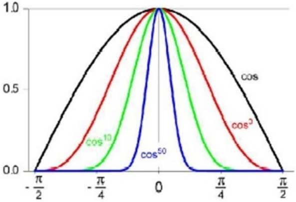
The *Gauss-BRDF* is defined as

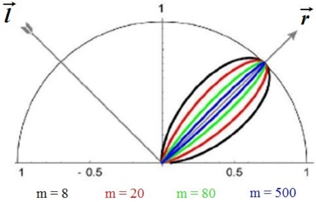
$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot \frac{\mathcal{N}(1 - \langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle \mid 0\,, \sigma^2)}{\cos(\theta_i)} \quad , \quad \boldsymbol{\kappa}_s \leq \boldsymbol{1} \quad , \quad \sigma^2 > 0$$

However, this BRDF has some major drawbacks:

- It is only suitable for small variances σ^2
- Normal distribution has no compact support $\rightarrow \forall x \colon \mathcal{N}(x \mid \mu\,, \sigma^2) > 0$
- Conservation of energy can not be guaranteed

The Phong-BRDF[6] was the first model proposed for specular materials. Specular surfaces reflect most of the incoming light around the ideal reflection direction, thus the BRDF should be maximum at this direction and should decrease sharply.





Definition (Phong-BRDF)

The *Phong-BRDF* is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m}{\cos(\theta_i)} = \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m}{\langle\, \boldsymbol{l} \,|\, \boldsymbol{n} \,\rangle}$$

[6] B. T. Phong. “Illumination for computer generated pictures”. In: *Communications of the ACM* 18.6 (1975), pp. 311–317.

The Phong-BRDF is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m}{\cos(\theta_i)} = \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m}{\langle\, \boldsymbol{l} \,|\, \boldsymbol{n} \,\rangle}$$

The factor $\boldsymbol{\kappa}_s$ is proportional to the Fresnel function, but is less than that, because now the surface is not an ideal mirror. It can be assumed to be independent of the wavelength and the incident angle for non-conducting materials (the highlight of a white lightsource on plastics is also white).

Problem: The original Phong model is not physically plausible since it is not reciprocal!

To prove that the Phong-BRDF is not reciprocal, we need a small lemma:

Lemma (Dot Products)

Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ *be a symmetric matrix and* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ *be some arbitrary vectors. Then the following holds:*

$$\langle\, \boldsymbol{A}\, \boldsymbol{x} \,|\, \boldsymbol{y} \,\rangle = \langle\, \boldsymbol{x} \,|\, \boldsymbol{A}\, \boldsymbol{y} \,\rangle$$

Proof.

Using the fact that the dot product can be written as a matrix multiplication, the associativity of the matrix multiplication and the symmetry of $\boldsymbol{A}$, we get

$$\langle\, \boldsymbol{A}\,\boldsymbol{x} \,|\, \boldsymbol{y} \,\rangle = (\boldsymbol{A}\,\boldsymbol{x})^T\,\boldsymbol{y} = \boldsymbol{x}^T\,\boldsymbol{A}^T\,\boldsymbol{y} = \boldsymbol{x}^T\,\boldsymbol{A}\,\boldsymbol{y} = \boldsymbol{x}^T\,(\boldsymbol{A}\,\boldsymbol{y}) = \langle\, \boldsymbol{x} \,|\, \boldsymbol{A}\,\boldsymbol{y} \,\rangle$$

□

Now, reciprocity means we can swap $\boldsymbol{l}$ and $\boldsymbol{v}$ and the BRDF remains the same. If we use the previous lemma and $\boldsymbol{r} = (\boldsymbol{I} - 2\,\boldsymbol{n}\,\boldsymbol{n}^T)\,\boldsymbol{l} =: \boldsymbol{H}\,\boldsymbol{l}$, we see that

$$\begin{aligned} \boldsymbol{f}_{BRDF}(\boldsymbol{l},\boldsymbol{v}) &= \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m}{\langle\, \boldsymbol{l} \,|\, \boldsymbol{n} \,\rangle} = \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{H}\,\boldsymbol{l} \,|\, \boldsymbol{v} \,\rangle^m}{\langle\, \boldsymbol{l} \,|\, \boldsymbol{n} \,\rangle} \\ &\neq \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{H}\,\boldsymbol{v} \,|\, \boldsymbol{l} \,\rangle^m}{\langle\, \boldsymbol{v} \,|\, \boldsymbol{n} \,\rangle} = \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{v} \,|\, \boldsymbol{H}\,\boldsymbol{l} \,\rangle^m}{\langle\, \boldsymbol{v} \,|\, \boldsymbol{n} \,\rangle} = \boldsymbol{\kappa}_s \cdot \frac{\langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m}{\langle\, \boldsymbol{v} \,|\, \boldsymbol{n} \,\rangle} = \boldsymbol{f}_{BRDF}(\boldsymbol{v},\boldsymbol{l}) \end{aligned}$$

since in general $\langle\, \boldsymbol{l} \,|\, \boldsymbol{n} \,\rangle \neq \langle\, \boldsymbol{v} \,|\, \boldsymbol{n} \,\rangle$. To make the Phong model reciprocal, we need to remove this factor.

To make the Phong model reciprocal, we need to remove this factor.

Definition (Reciprocal Phong-BRDF)

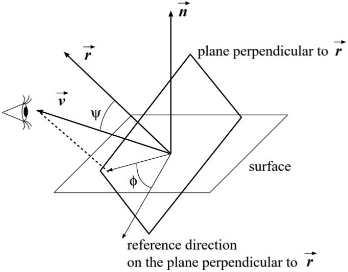
The Reciprocal Phong-BRDF is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot \langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m$$

Now, we want to make this BRDF physically plausible. So we have to compute the albedo.

Unfortunately, the domain of (ψ, ϕ) with $\psi := \langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle$ is rather complicated

- Those pairs that would identify a direction pointing into the object should be excluded from reflection directions
- Without this exclusion the albedo is over-estimated



Idea: Define a function $Cut(\theta_i)$ which models this effect to get a correct estimate of the albedo:

$$\boldsymbol{a}(\boldsymbol{l}) = Cut(\theta_i) \int_{\phi_o=0}^{2\pi} \int_{\psi=0}^{\pi/2} \boldsymbol{\kappa}_s \cdot \cos(\psi)^m \cdot \cos(\theta_o) \sin(\psi) \, d\psi \, d\phi_o$$

The function $Cut(\theta_i)$ is 1 at perpendicular illumination (no over-estimation) and decreases to a $\frac{1}{2}$ at horizontal illumination (worst over-estimation). When the illumination is perpendicular, then $\psi = \theta_o$, thus the albedo for this case is

$$\begin{aligned}
\boldsymbol{a}(\boldsymbol{l}_0) &= Cut(0) \int_{\phi_o=0}^{2\pi} \int_{\psi=0}^{\pi/2} \boldsymbol{\kappa}_s \cdot \cos(\psi)^m \cdot \cos(\theta_o) \sin(\psi) \, d\psi \, d\phi_o \\
&= 1 \cdot \int_{\phi_o=0}^{2\pi} \int_{\psi=0}^{\pi/2} \boldsymbol{\kappa}_s \cdot \cos(\psi)^{m+1} \sin(\psi) \, d\psi \, d\phi_o \\
&= \boldsymbol{\kappa}_s \cdot \frac{2\pi}{m+2}
\end{aligned}$$

So if we scale the reciprocal Phong-BRDF and limit $\boldsymbol{\kappa}_s$, then it is physically plausible.

So if we scale the reciprocal Phong-BRDF and limit $\boldsymbol{\kappa}_s$, then it is physically plausible.

Definition (Cosine-Lobe-BRDF[7])

The *Cosine-Lobe-BRDF* is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot \frac{m+2}{2\pi} \langle \boldsymbol{r} \,|\, \boldsymbol{v} \rangle^m \quad , \quad \boldsymbol{\kappa}_s \leq \mathbf{1}$$

Now, one might ask, how this model can be generalized. Generalization means to construct a BRDF similar to the Cosine-Lobe-BRDF but with a lobe that is not centered at the ideal reflection direction.

$$\begin{aligned} \boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) &= \boldsymbol{\kappa}_s \cdot \frac{m+2}{2\pi} \langle \boldsymbol{r} \,|\, \boldsymbol{v} \rangle^m = \boldsymbol{\kappa}_s \cdot \frac{m+2}{2\pi} (\boldsymbol{r}^T \boldsymbol{v})^m \\ &= \boldsymbol{\kappa}_s \cdot C_s \cdot (\boldsymbol{l}^T \boldsymbol{H} \boldsymbol{v})^m =: \boldsymbol{\kappa}_s \cdot (\boldsymbol{l}^T \boldsymbol{H}_s \boldsymbol{v})^m \end{aligned}$$

Here, $\boldsymbol{H}_s = C_s \cdot \boldsymbol{H}$ and $C_s = \frac{m+2}{2\pi}$ where $\boldsymbol{H}$ is the Householder matrix. We can generalize this equation by replacing the scaled Householder matrix $\boldsymbol{H}_s$ by some arbitrary symmetric matrix $\boldsymbol{M}$.

[7] R. R. Lewis. "Making shaders more physically plausible". In: *Computer Graphics Forum*. Vol. 13. 2. 1994, pp. 109–120.

We can generalize this equation by replacing the scaled Householder matrix $\boldsymbol{H}_s$ by some arbitrary symmetric matrix $\boldsymbol{M}$.

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l},\boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot (\boldsymbol{l}^T\,\boldsymbol{M}\,\boldsymbol{v})^m$$

What remains is the choice of $\boldsymbol{M}$. Given a singular value decomposition of it ($\boldsymbol{M} = \boldsymbol{Q}^T\boldsymbol{D}\boldsymbol{Q}$), we obtain

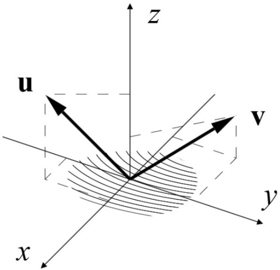
$$\boldsymbol{f}_{BRDF}(\boldsymbol{l},\boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot (\boldsymbol{l}^T\,\boldsymbol{M}\,\boldsymbol{v})^m = \boldsymbol{\kappa}_s \cdot (\boldsymbol{l}^T\,\boldsymbol{Q}^T\,\boldsymbol{D}\,\boldsymbol{Q}\,\boldsymbol{v})^m = \boldsymbol{\kappa}_s \cdot (\boldsymbol{l}^{'T}\,\boldsymbol{D}\,\boldsymbol{v}^{'})^m$$

Thus, generalization means that we transform the incoming and outgoing direction to a local coordinate system

$$\boldsymbol{l}^{'} = \boldsymbol{Q}\,\boldsymbol{l} \quad , \quad \boldsymbol{v}^{'} = \boldsymbol{Q}\,\boldsymbol{v}$$

and then compute a scaled dot product.

$$\boldsymbol{l}^{'T}\boldsymbol{D}\boldsymbol{v}^{'} = l_x^{'}\,D_x\,v_x^{'} + l_y^{'}\,D_y\,v_y^{'} + l_z^{'}\,D_z\,v_z^{'}$$

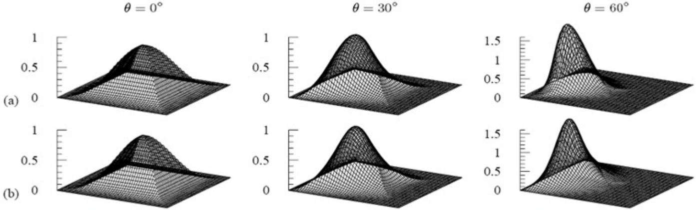


Definition (Lafortune-BRDF[8])

The Lafortune-BRDF is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \boldsymbol{\kappa}_s \cdot (l_x^{'} \, C_x \, v_x^{'} + l_y^{'} \, C_y \, v_y^{'} + l_z^{'} \, C_z \, v_z^{'})^m \quad , \quad \boldsymbol{\kappa}_s \leq \mathbf{1}$$

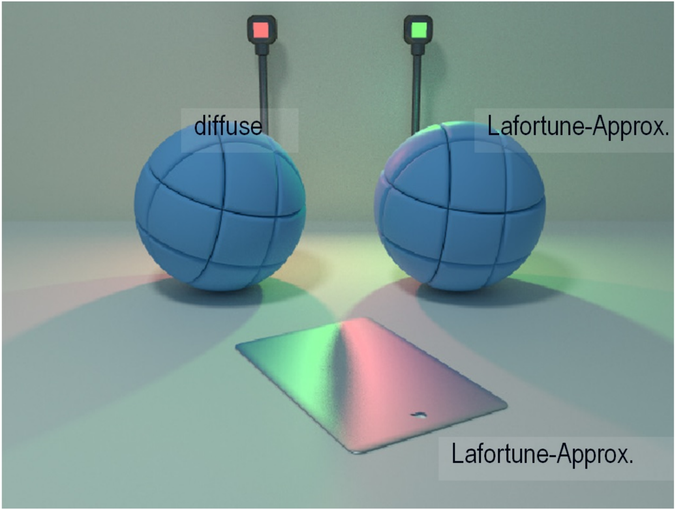
Measured BRDF's can often be approximated well as a sum of Lafortune lobes (nonlinear optimization):



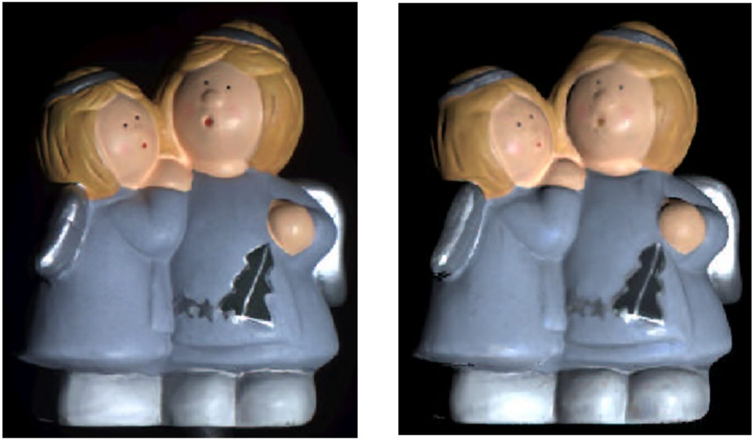
Brushed aluminum: **a)** Measured BRDF **b)** fitted BRDF [Lafortune 97]

[8] E. P. Lafortune et al. "Non-linear approximation of reflectance functions". In: *Proceedings of the 24th annual conference on Computer graphics and interactive techniques*. ACM Press/Addison-Wesley Publishing Co. 1997, pp. 117–126.

Results:

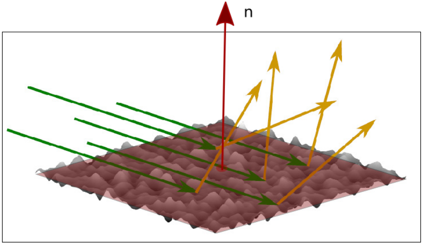


Results:



Left : Photography , **Right** : Reconstructed model under similar light conditions.
9 different BRDF's were used to describe the material.

Idea: Reflection of light on a can be more rigorously analyzed by modeling surface irregularities at microscale as has been first proposed by Torrance, Sparrow[9], and introduced computer graphics by Cook and Torrance[10].
At the macroscopic scale we assume a planar geometric surface with surface normal n, whose interface is, at the microscopic scale, rough and composed of **microfacets**.



This representation is used to explain and make predictions about scattering events occurring at the geometric surface interface and derive the corresponding BSDF models.

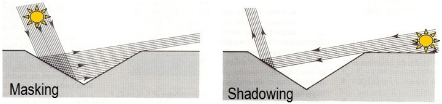
[9] K. E. Torrance and E. M. Sparrow. "Theory for off-specular reflection from roughened surfaces". In: *JOSA* 57.9 (1967), pp. 1105–1112.
[10] R. L. Cook. "A Reflectance Model for Computer Graphics". en. In: *ACM Transactions on Graphics* 1.1 (), p. 18.

The the probability density of the transfer function, i.e.

$$\boldsymbol{w}(\boldsymbol{\omega}_i, \boldsymbol{x}, \boldsymbol{\omega}_o)\, d\omega_o = \Pr(\text{photon gets scattered to } d\omega_o \mid \text{photon is coming from } \boldsymbol{\omega}_i)$$

depends on the following events:

1. **Orientation of hit microfacet**: e.g. in case of perfect specular micro facets the hit microfacet must have the normal $\boldsymbol{h} = \frac{\boldsymbol{\omega}_i + \boldsymbol{\omega}_o}{\|\boldsymbol{\omega}_i + \boldsymbol{\omega}_o\|_2}$
2. **Shadowing or masking**: The given microfacet is not hidden by other microfacets with respect to the incoming light, and the reflected photon is not onto another microfacet



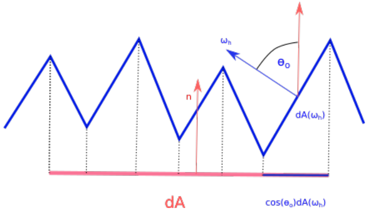
3. **Reflection behavior on mircrofacet**: The photon is not absorbed by the microfacet, e.g. if it is supposed to be a perfect mirror

Since these events are believed to be stochastically independent, their probabilities can be calculated independently.

$$\boldsymbol{w}(\boldsymbol{\omega}_i, \boldsymbol{\omega}_o)\, d\omega_o = \Pr(\text{orientation}) \cdot \Pr(\text{no shadowing or masking}) \cdot \Pr(\text{reflection})$$

Microfacet distribution functions

The distribution function of microfacet orientations $D(\boldsymbol{\omega}_h)$ gives the ratio of the area of microfacets with the surface normal $\boldsymbol{\omega}_h$ and the total area of the differential surface element dA.



Microfacet distribution functions must be normalized to ensure that they are physically plausible.

Given a differential area of the microsurface $dA(\boldsymbol{\omega}_h)$, the projected area of the microfacet faces above that area must be equal to

$$\int_{\Omega} D(\boldsymbol{\omega}_h) \cos(\theta_h) d\boldsymbol{\omega}_h = 1.$$

What are suitable Normal Distribution Functions (NDFs) $D(\boldsymbol{h})$?

- Blinn (1977): Gaussian distribution

$$D(\boldsymbol{h}) = \mathrm{const} \cdot e^{-\frac{\alpha^2}{m^2}}$$

  with α being the angle between $\boldsymbol{n}$ and $\boldsymbol{h}$ and m a roughness parameter

- Ashikmin et al. (2000): User-specific distribution as a texture

- Beckmann and Spizzichino (1963): Isotropic Beckmann-Spizzichino model.

$$D(\boldsymbol{h}) = \frac{1}{\pi\,\alpha^2 \cos(\theta_h)^4} \cdot e^{-\frac{\tan(\theta_h)^2}{\alpha^2}}$$

  If σ is the RMS slope of the microfacets, then $\alpha = \sqrt{2}\sigma$.

- Anisotropic Beckmann-Spizzichino model.

$$D(\boldsymbol{h}) = \frac{1}{\pi\,\alpha_x \alpha_y \cos(\theta_h)^4} \cdot e^{-\tan(\theta_h)^2 \left(\frac{\cos(\Phi_h)^2}{\alpha_x^2} + \frac{\sin(\Phi_h)^2}{\alpha_y^2}\right)}$$

  In case that $\alpha_x = \alpha_y$ we get the isotropic variant of the Beckmann-Spizzichino model.

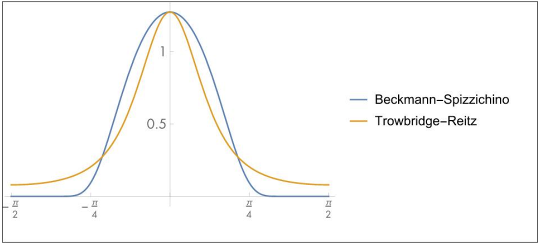


Figure: Graphs of isotropic Beckmann–Spizzichino and Trowbridge–Reitz microfacet distribution functions as a function of θ for $\alpha = 0.5$. Note that Trowbridge–Reitz has higher tails for values of with larger magnitudes.

- Trowbridge and Reitz (1975): Trowbridge Reitz model

$$D(\boldsymbol{h}) = \frac{1}{\pi\, \alpha_x \alpha_y \cos(\theta_h)^4 \left(1 + \tan(\theta_h)^2 (\cos(\Phi_h)^2/\alpha_x^2 + \sin(\Phi_h)^2/\alpha_y^2)\right)^2}$$

The model was independently derived by Walter et al. Walter et al. (2007), who dubbed it **"GGX"**.

The surface model of Heitz For validation purposes explicit instances of microflake models are of interest Heitz (2015),Ribardière et al. (2019).
For $i = 1, \dots, N$ compute 3 random variables from independent uniform random numbers U_i in the following way

$$\begin{aligned} \varPhi_i &= 2\pi U_1 \\ f_i^x &= \sqrt{-\log U_2} \cdot \cos(2\pi U_3) \cdot a_x \\ f_i^y &= \sqrt{-\log U_3} \cdot \cos(2\pi U_3) \cdot a_y \end{aligned}$$

where a_x, a_y are user defined parameters of the Beckmann-Spizzichino model and the the $U_i, i = 1, 2, 3$ are independent uniform random numbers.
Define the surface by

$$h(x, y) = \sqrt{\frac{2}{N}} \sum_{i=1}^{N} \cos(\varPhi_i + x f_i^x + y f_i^y)$$

The values of h are distributed according to a Gaussian distribution of RMS $\sigma_h = 1$.

The slopes of $h(x,y)$ are given by

$$\left(\frac{\partial h}{\partial x},\frac{\partial h}{\partial y}\right)=\sqrt{\frac{2}{N}}\sum_{i=1}^{N}(-f_i^x\sin(\Phi_i+xf_i^x+yf_i^y),-f_i^y\sin(\Phi_i+xf_i^x+yf_i^y))$$

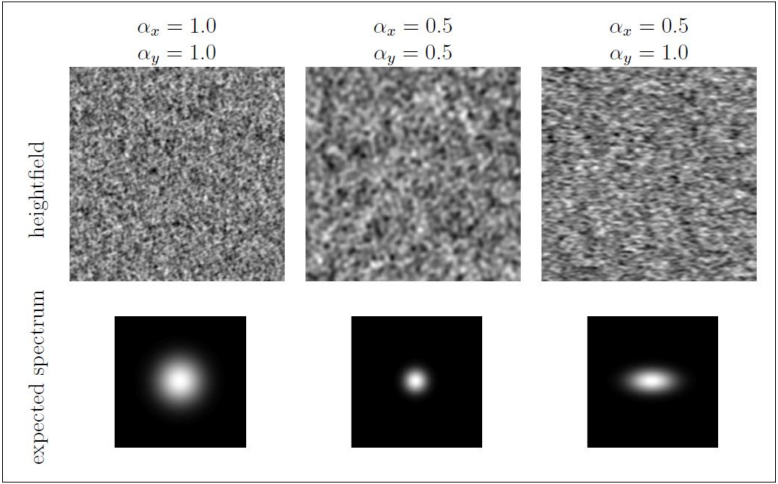
They are distributed according to a according to a Beckmann-Spizzichino distribution

with parameters $\sigma_x=\frac{a_x}{\sqrt{2}},\sigma_y=\frac{a_y}{\sqrt{2}}$.

The normal is given by

$$\omega_h(x,y)=\frac{(-\partial h/\partial x,-\partial h/\partial y,1)}{\sqrt{(\partial h/\partial x)^2+(\partial h/\partial y)^2+1}}$$

Examples


$\alpha_x = 1.0$
$\alpha_y = 1.0$
$\alpha_x = 0.5$
$\alpha_y = 0.5$
$\alpha_x = 0.5$
$\alpha_y = 1.0$
heightfield
expected spectrum

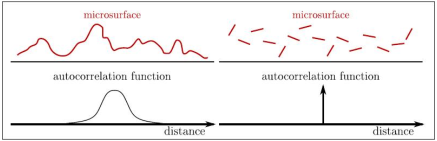
Shadowing and masking

In order to determine the probability that a point on the surface is both visible and illuminated (also known as the bistatic shadowing function) a set of approximations is made to make the problem tractable.

The most popular shadowing functions currently used are

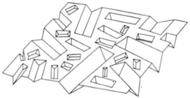
- modifications of those of Smith[11].

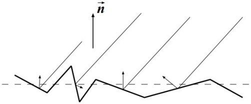
It assumes that the microsurface is not autocorrelated. This implies a random set of microfacets where the heights and the normals of the microsurface are independent random variables



- the original Torrance and Sparrow shadowing term[12].

It assumes an inconsistent model of an isotropic surface exclusively made by randomly oriented very long V-cavities.





[11] B. Smith. "Geometrical shadowing of a random rough surface". In: *IEEE transactions on antennas and propagation* 15.5 (1967), pp. 668–671.

[12] K. E. Torrance and E. M. Sparrow. "Theory for off-specular reflection from roughened surfaces". In: *JOSA* 57.9 (1967), pp. 1105–1112.

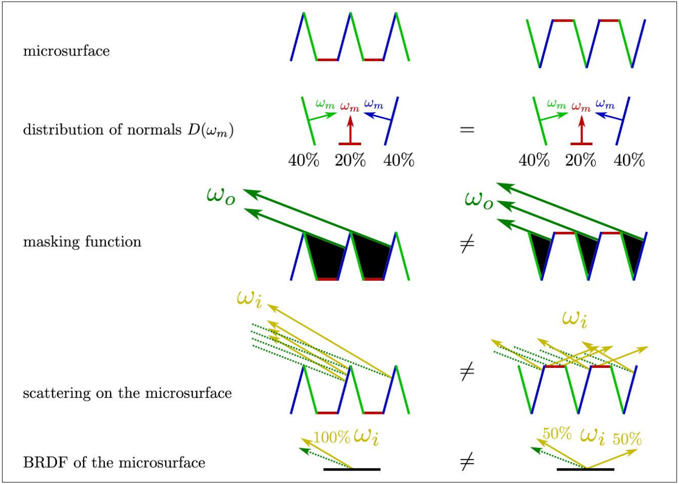
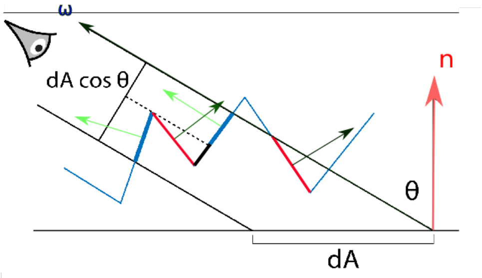


Figure: Microsurfaces with the same distribution of normals but with different profiles result in different BRDFs[13].

[13] E. Heitz. "Understanding the masking-shadowing function in microfacet-based BRDFs". In: (2014).



- Backfacing microfacets with respect to $\boldsymbol{\omega}$ will not be visible. The fraction of microfacets with normal $\boldsymbol{\omega}_h$ that are visible from direction $\boldsymbol{\omega}$ are given by the Smith's masking function $G_1(\boldsymbol{\omega}, \boldsymbol{\omega}_h)$.
- A differential area dA on the surface has area $dA\cos(\theta)$ when viewed from a direction $\boldsymbol{\omega}$ that makes an angle θ with the surface normal. This leads to a normalization constraint for G_1

$$\cos(\theta) = \int_{\Omega} G_1(\boldsymbol{\omega}, \boldsymbol{\omega}_h) \max(0, \langle \boldsymbol{\omega} \,|\, \boldsymbol{\omega}_h \rangle) D(\boldsymbol{\omega}_h) d\boldsymbol{\omega}_h$$

If the probability that a microfacet is visible is independent of its orientation the masking-shadowing function does not depent on $\boldsymbol{\omega}_h$, G_1 depends on $\boldsymbol{\omega}$ only, i.e. $G_1(\boldsymbol{\omega}, \boldsymbol{\omega}_h) = G_1(\boldsymbol{\omega})$
Let $A^+(\boldsymbol{\omega})$ be the projected area of the forward-facing microfacet as seen from $\boldsymbol{\omega}$ and $A^-(\boldsymbol{\omega})$ be the projected area of the backward-facing microfacets.
Then

$$\cos(\theta) = A^+(\boldsymbol{\omega}) - A^-(\boldsymbol{\omega})$$

Therefore, the masking shadowing function can be expressed as the ratio of visible microfacet area to total forward-facing microfacet area:

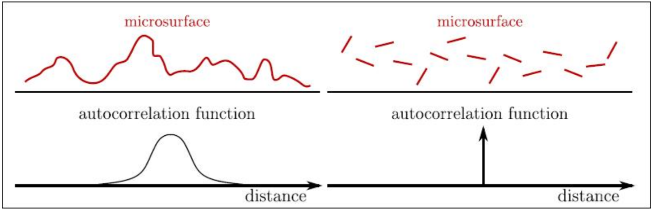
$$G_1(\boldsymbol{\omega}) = \frac{A^+(\boldsymbol{\omega}) - A^-(\boldsymbol{\omega})}{A^+(\boldsymbol{\omega})}.$$

Shadowing-masking functions are traditionally expressed in terms of an auxiliary function $\Lambda(\boldsymbol{\omega})$, which measures invisible masked microfacet area per visible microfacet area:

$$\Lambda(\boldsymbol{\omega}) = \frac{A^-(\boldsymbol{\omega})}{A^+(\boldsymbol{\omega}) - A^-(\boldsymbol{\omega})} = \frac{A^-(\boldsymbol{\omega})}{\cos(\theta)}.$$

Some algebra lets us express $G_1(\boldsymbol{\omega})$ in terms of $\Lambda(\boldsymbol{\omega})$:

$$G_1(\boldsymbol{\omega}) = \frac{1}{1 + \Lambda(\boldsymbol{\omega})}.$$



uncorrelated height assumption Under the assumption of no correlation of the heights of nearby points, $\Lambda(\boldsymbol{\omega})$ is given by

- Beckmann-Spizzichino distribution

$$\Lambda(\boldsymbol{\omega}) = \frac{1}{2}\left(erf(a) - 1 + \frac{e^{-a^2}}{a\sqrt{\pi}}\right),$$

- Trowbridge-Reitz distribution

$$\Lambda(\boldsymbol{\omega}) = \frac{-1 + \sqrt{1 + \alpha^2 \tan(\theta)^2}}{2}$$

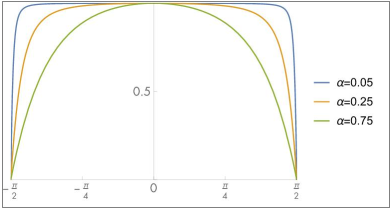


Figure: The Masking-Shadowing Function $G_1(\omega)$ for the Trowbridge-Reitz Distribution.

To get shadowing functions $G(\omega_o, \omega_i)$ that depend on both, ω_o and ω_i a practical approach is to assume that shadowing and masking are independent. In this case we get

$$G(\omega_o, \omega_i) = G_1(\omega_o) \cdot G_1(\omega_i).$$

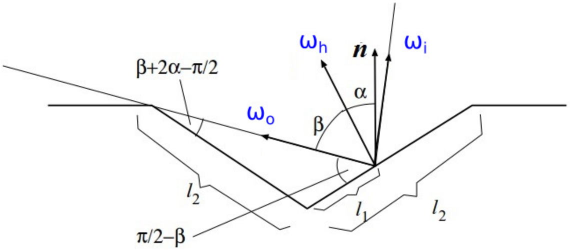
In practice, however, these probabilities aren't independent, and this formulation underestimates G, which can easily be seen in the case where $\omega_o = \omega_i$.
A more accurate model can be derived assuming that microfacet visibility is more likely the higher up a given point on a microfacet is. This assumption leads to the model

$$G(\omega_o, \omega_i) = \frac{1}{1 + \Lambda(\omega_o) + \Lambda(\omega_i)}.$$

To derive the original Cook-Torrance shadowing function we assume that all microfacets have equal area f.
We can easily recognize that the probability of masking is

$$\Pr(\text{masking}) = \frac{l_1}{l_2}$$

where l_2 is the one-dimensional length of the microfacet, and l_1 describes the boundary case from where the beam is masked.



The angles of the triangle formed by the bottom of the microfacet wedge and the beam in the boundary case can be expressed by the angles $\alpha = \cos(\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_h} \,\rangle)^{-1}$ and $\beta = \cos(\langle\, \boldsymbol{\omega_o} \,|\, \boldsymbol{\omega_h} \,\rangle)^{-1}$ by geometric considerations and by using the law of reflection.

Applying the sine law for this triangle, we get

$$\mathrm{Pr}(\textsf{no masking}) = 1 - \frac{l_1}{l_2} = 1 - \frac{\sin(\beta + 2\alpha - \pi/2)}{\sin(\pi/2 - \beta)}$$

Using the addition theorem for the sine, we get

$$\sin(\pi/2 - \beta) = \sin(\pi/2)\cos(\beta) - \cos(\pi/2)\sin(\beta) = \cos(\beta)$$
$$\sin(\beta + 2\alpha - \pi/2) = \sin(\beta + 2\alpha)\cos(\pi/2) - \cos(\beta + 2\alpha)\sin(\pi/2) = -\cos(\beta + 2\alpha)$$

which leads to

$$\begin{aligned}\mathrm{Pr}(\textsf{no masking}) &= 1 - \frac{\sin(\beta + 2\alpha - \pi/2)}{\sin(\pi/2 - \beta)} \\ &= 1 + \frac{\cos(\beta + 2\alpha)}{\cos(\beta)} \\ &= \frac{\cos(\beta) + \cos(\beta + 2\alpha)}{\cos(\beta)}\end{aligned}$$

Now, we split the cosine using the addition theorem for it.

Now, we split the cosine using the addition theorem for it.

$$\begin{aligned}\cos(\beta) + \cos(\beta + 2\alpha) &= \cos(\beta) + \cos(\beta + \alpha)\cos(\alpha) - \sin(\beta + \alpha)\sin(\alpha) \\ &= \cos(\beta + \alpha)\cos(\alpha) + \cos(\beta) - \sin(\beta + \alpha)\sin(\alpha)\end{aligned}$$

The last part can further be refined to

$$\begin{aligned}\cos(\beta) - \sin(\beta + \alpha)\sin(\alpha) &= \cos(\beta) - \sin(\beta)\cos(\alpha)\sin(\alpha) - \cos(\beta)\sin(\alpha)^2 \\ &= \cos(\beta)\,(1 - \sin(\alpha)^2) - \sin(\beta)\cos(\alpha)\sin(\alpha) \\ &= \cos(\beta)\cos(\alpha)^2 - \sin(\beta)\cos(\alpha)\sin(\alpha) \\ &= (\cos(\beta)\cos(\alpha) - \sin(\beta)\sin(\alpha))\cos(\alpha) \\ &= \cos(\beta + \alpha)\cos(\alpha)\end{aligned}$$

which leads to

$$\cos(\beta) + \cos(\beta + 2\alpha) = 2\cos(\beta + \alpha)\cos(\alpha)$$

The final result is

$$\mathrm{Pr}(\text{no masking}) = \frac{\cos(\beta) + \cos(\beta + 2\alpha)}{\cos(\beta)} = 2\,\frac{\cos(\alpha)\cdot\cos(\beta + \alpha)}{\cos(\beta)}$$

The final result is

$$\mathrm{Pr}(\text{no masking}) = \frac{\cos(\beta) + \cos(\beta + 2\alpha)}{\cos(\beta)} = 2\,\frac{\cos(\alpha)\cdot\cos(\beta+\alpha)}{\cos(\beta)}$$

According to the definition of the angles $\alpha = \cos(\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_h} \,\rangle)^{-1}$ and $\beta = \cos(\langle\, \boldsymbol{\omega_o} \,|\, \boldsymbol{\omega_h} \,\rangle)^{-1}$, the probability is given by

$$\mathrm{Pr}(\text{no masking}) = \min\left\{2\cdot\frac{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_h} \,\rangle \cdot \langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_o} \,\rangle}{\langle\, \boldsymbol{\omega_o} \,|\, \boldsymbol{\omega_h} \,\rangle}\,,\, 1\right\}$$

since it is limited to 1 if there is no occlusion.

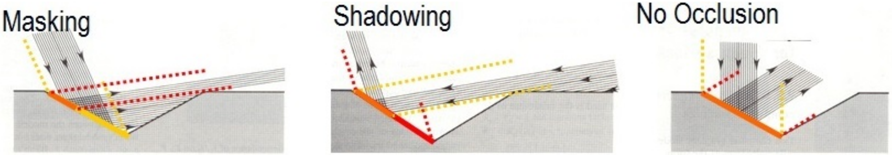
The probability of shadowing can be derived in exactly the same way, only $\boldsymbol{l}$ should be substituted for $\boldsymbol{v}$:

$$\mathrm{Pr}(\text{no shadowing}) = \min\left\{2\cdot\frac{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_h} \,\rangle \cdot \langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle}{\langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle}\,,\, 1\right\}$$

The probability of neither shadowing nor masking taking place can be approximated by the minimum of the two probabilities.

The probability of neither shadowing nor masking taking place can be approximated by the minimum of the two probabilities.

$$\Pr(\text{no shadowing or masking}) = \min\left\{2 \cdot \frac{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_h} \,\rangle \cdot \langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle}{\langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle}\,,\, 2 \cdot \frac{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_h} \,\rangle \cdot \langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_o} \,\rangle}{\langle\, \boldsymbol{\omega_o} \,|\, \boldsymbol{\omega_h} \,\rangle}\,,\, 1\right\}$$
$$=: G(\boldsymbol{n}, \boldsymbol{\omega_i}, \boldsymbol{\omega_o})$$



Note: This approximation upper-bounds the real probability, particularly at grazing angles, which can result in the violation of the energy balance.

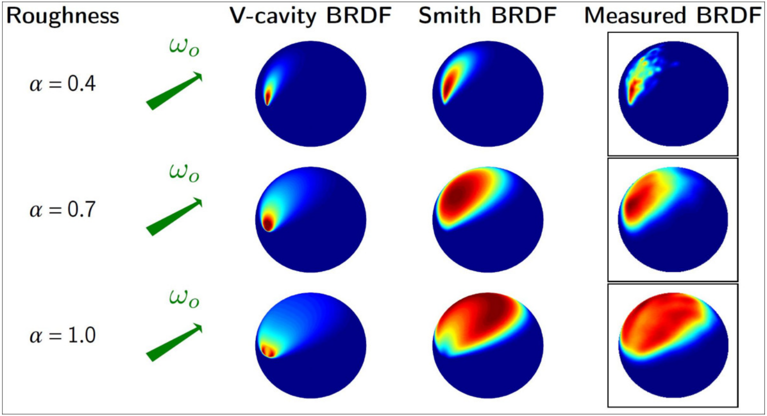


Figure: The BRDFs associated with an isotropic Beckmann distribution and different masking functions at grazing incidence ($\theta_o = 1.5$) Measured BRDF computed with Monte Carlo raytracing on a procedural surface with Gaussian statistics parametrized by the roughness parameter α[14].

[14] E. Heitz. "Understanding the masking-shadowing function in microfacet-based BRDFs". In: (2014).

Step 1: Now, we are interested in the probability

$$\Pr(\text{orientation})$$

First, let us examine a surface element dA. If a photon arrives from direction $\boldsymbol{l}$ to a surface element dA, the visible area of the surface element will be

$$dA \, \langle \, \boldsymbol{n} \, | \, \boldsymbol{l} \, \rangle$$

while the total visible area of the microfacets having their normal in the direction around $\boldsymbol{h}$ will be the product of the visible size of a single microfacet

$$f \, \langle \, \boldsymbol{h} \, | \, \boldsymbol{l} \, \rangle$$

and the number of properly oriented microfacets. The number of properly oriented microfacets, in turn, is the product of the probability that a microfacet is properly oriented

$$D(\boldsymbol{h}) \, d\omega_h$$

and the number of all microfacets in the surface element

$$dA/f$$

The probability of finding an appropriate microfacet aligned with the photon is the visible size of properly oriented microfacets divided by the visible surface area, thus we obtain

$$\mathrm{Pr}(\text{orientation}) = \frac{f\,\langle\,\boldsymbol{\omega_h}\,|\,\boldsymbol{\omega_i}\,\rangle \cdot D(\boldsymbol{\omega_h})\,d\omega_h \cdot dA/f}{dA\,\langle\,\boldsymbol{n}\,|\,\boldsymbol{\omega_i}\,\rangle} = \frac{D(\boldsymbol{\omega_h})\,d\omega_h\,\langle\,\boldsymbol{\omega_h}\,|\,\boldsymbol{\omega_i}\,\rangle}{\langle\,\boldsymbol{n}\,|\,\boldsymbol{\omega_i}\,\rangle}$$

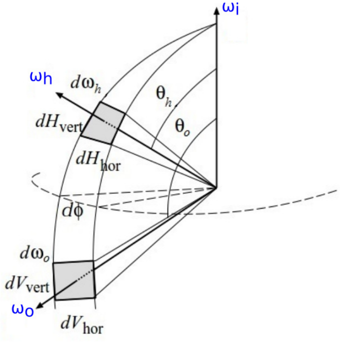
Unfortunately, in the BRDF the transfer probability density uses $d\omega_o$ instead of $d\omega_h$, thus we have to determine $d\omega_h$.

Defining a spherical coordinate system (θ, ϕ), with the north pole in the direction of $\boldsymbol{\omega_i}$, the solid angles are expressed by the product of vertical and horizontal arcs

$$d\omega_o = dV_{hor}\,dV_{vert}\ ,\ \ d\omega_h = dH_{hor}\,dH_{vert}$$

By using geometric considerations and applying the law of reflection, we get

$$dV_{hor} = d\phi\sin(\theta_o)\ ,\ \ dH_{hor} = d\phi\sin(\theta_h)$$
$$dV_{vert} = 2\,dH_{vert}$$



This in turn yields

$$\frac{d\omega_h}{d\omega_o} = \frac{dH_{hor}\, dH_{vert}}{dV_{hor}\, dV_{vert}} = \frac{d\phi \sin(\theta_h)\, dH_{vert}}{d\phi \sin(\theta_o)\, 2\, dH_{vert}} = \frac{\sin(\theta_h)}{2\sin(\theta_o)}$$

Since, $\theta_o = 2\,\theta_h$, we get

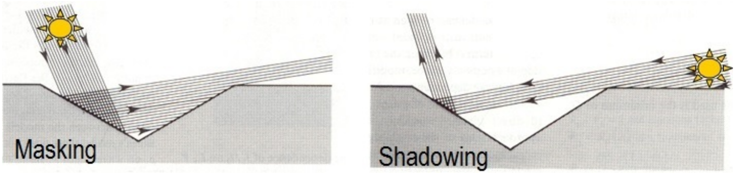
$$\frac{d\omega_h}{d\omega_o} = \frac{\sin(\theta_h)}{2\sin(2\,\theta_h)} = \frac{\sin(\theta_h)}{2\,(2\sin(\theta_h)\cos(\theta_h))} = \frac{1}{4\cos(\theta_h)} = \frac{1}{4\,\langle\, \boldsymbol{\omega_h} \,|\, \boldsymbol{\omega_i} \,\rangle}$$

Thus the orientation probability is

$$\begin{aligned}
\mathrm{Pr}(\text{orientation}) &= \frac{D(\boldsymbol{\omega_h})\, d\omega_h \,\langle\, \boldsymbol{\omega_h} \,|\, \boldsymbol{\omega_i} \,\rangle}{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle} \\
&= \frac{D(\boldsymbol{\omega_h})\,\langle\, \boldsymbol{\omega_h} \,|\, \boldsymbol{\omega_i} \,\rangle}{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle} \frac{d\omega_h}{d\omega_o} d\omega_o \\
&= \frac{D(\boldsymbol{\omega_h})\,\langle\, \boldsymbol{\omega_h} \,|\, \boldsymbol{\omega_i} \,\rangle}{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle} \frac{1}{4\,\langle\, \boldsymbol{\omega_h} \,|\, \boldsymbol{\omega_i} \,\rangle}\, d\omega_o \\
&= \frac{D(\boldsymbol{\omega_h})}{4\,\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle}\, d\omega_o
\end{aligned}$$

Step 2: Next, we are interested in the probability[15]

$$\Pr(\text{no shadowing or masking})$$



- The visibility of the microfacets from direction $\boldsymbol{v}$ means that the reflected photon does not run into another microfacet. The collision is often referred to as masking.
- Similarly, shadowing describes the effect that only a part of the microfacet is illuminated and therefore photons are colliding the other way around.

Note: Another effect is inter-reflection between microfacets. This has to be modeled separately and is skipped here.

[15] K. E. Torrance and E. M. Sparrow. "Theory for off-specular reflection from roughened surfaces". In: *JOSA* 57.9 (1967), pp. 1105–1112.

Step 3: Last, we are interested in the probability

$$\mathrm{Pr}(\text{reflection}) = F(\lambda, \langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle)$$

which is described by the Fresnel Equations. It depends on index of refraction n, the extinction coefficient k of the surface and the angle of illumination $\langle\, \boldsymbol{l} \,|\, \boldsymbol{h} \,\rangle$.

Problem: In general, both n and k vary with wavelength, but their values are not known. On the other hand, experimentally measured values of the reflectance at normal incidence are frequently known.

Idea: Fit the Fresnel equation to the measured normal reflectance F_0:

$$F(\lambda, \langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle) = \frac{1}{2} \frac{(g-c)^2}{(g+c)^2} \left(1 + \frac{(c\,(g+c)-1)^2}{(c\,(g-c)+1)^2}\right)$$

where

$$c = \langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle \quad , \quad g^2 = n^2 + c^2 - 1$$

For non-metals with $k = 0$ and normal incidence $\boldsymbol{l} = \boldsymbol{h}$, we get $c = 1$ and $g = n$, so the Fresnel term reduces to

$$\begin{aligned}
F_0(\lambda) &= \frac{1}{2}\frac{(n-1)^2}{(n+1)^2}\left(1 + \frac{(1\,(n+1)-1)^2}{(1\,(n-1)+1)^2}\right) \\
&= \frac{1}{2}\frac{(n-1)^2}{(n+1)^2}\left(1 + \frac{n^2}{n^2}\right) \\
&= \frac{(n-1)^2}{(n+1)^2}
\end{aligned}$$

Solving for n leads to

$$\begin{aligned}
F_0(\lambda) &= \frac{(n-1)^2}{(n+1)^2} \\
\sqrt{F_0(\lambda)} &= \frac{n-1}{n+1} \qquad \text{since } F_0(\lambda) \geq 0 \text{ and } n \geq 1 \\
\sqrt{F_0(\lambda)}\,n + \sqrt{F_0(\lambda)} &= n - 1 \\
(\sqrt{F_0(\lambda)} - 1)\,n &= -\sqrt{F_0(\lambda)} - 1
\end{aligned}$$

$$
\begin{aligned}
(\sqrt{F_0(\lambda)} - 1)\, n &= -\sqrt{F_0(\lambda)} - 1 \\
(1 - \sqrt{F_0(\lambda)})\, n &= 1 + \sqrt{F_0(\lambda)} \\
n(\lambda) &= \frac{1 + \sqrt{F_0(\lambda)}}{1 - \sqrt{F_0(\lambda)}}
\end{aligned}
$$

Now, one has to do this for each wavelength. Another simplification is to assume that the refraction indices $n(\lambda)$ are almost the same for each wavelength (and compensate this with $\boldsymbol{\kappa}_s$), so we need only one measure for F_0. This leads to the approximation

$$
F(\lambda, \langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle) \approx \boldsymbol{\kappa}_s \cdot F(\langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle)
$$

with

$$
F(\langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle) = \frac{1}{2} \frac{(g-c)^2}{(g+c)^2} \left(1 + \frac{(c\,(g+c) - 1)^2}{(c\,(g-c) + 1)^2} \right)
$$

and the given parameters

$$
c = \langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle \quad , \quad g^2 = n^2 + c^2 - 1 \quad , \quad n = \frac{1 + \sqrt{F_0}}{1 - \sqrt{F_0}}
$$

Conclusion: Now we can summarize the results by multiplying the probabilities of the independent events to express

$$\begin{aligned}\boldsymbol{w}(\boldsymbol{\omega_i}, \boldsymbol{\omega_o})\, d\omega_o &= \Pr(\text{orientation}) \cdot \Pr(\text{no shadowing or masking}) \cdot \Pr(\text{reflection}) \\ &= \frac{D(\boldsymbol{\omega_h})}{4\,\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle}\, d\omega_o \cdot G(\boldsymbol{n}, \boldsymbol{\omega_i}, \boldsymbol{\omega_o}) \cdot \boldsymbol{\kappa}_s \cdot F(\langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle)\end{aligned}$$

A BRDF is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{\omega_i}, \boldsymbol{\omega_o}) = \frac{\boldsymbol{w}(\boldsymbol{\omega_i}, \boldsymbol{\omega_o})}{\cos(\theta_o)} = \frac{\boldsymbol{w}(\boldsymbol{\omega_i}, \boldsymbol{\omega_o})}{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_o} \,\rangle}$$

Definition (Cook-Torrance-BRDF[16])

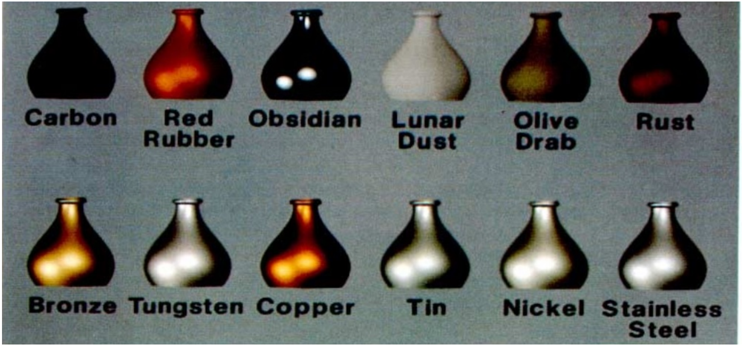
The *Cook-Torrance-BRDF* is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{\omega_i}, \boldsymbol{\omega_o}) = \frac{\boldsymbol{\kappa}_s}{4}\, \frac{D(\boldsymbol{\omega_h}) \cdot G(\boldsymbol{n}, \boldsymbol{\omega_i}, \boldsymbol{\omega_o}) \cdot F(\langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle)}{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle\, \langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_o} \,\rangle} \quad , \quad \boldsymbol{\kappa}_s \leq \mathbf{1}$$

[16] R. L. Cook and K. E. Torrance. “A reflectance model for computer graphics”. In: *ACM Transactions on Graphics (TOG)* 1.1 (1982), pp. 7–24.

The Cook-Torrance-BRDF is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{\omega_i}, \boldsymbol{\omega_o}) = \frac{\boldsymbol{\kappa}_s}{4} \frac{D(\boldsymbol{\omega_h}) \cdot G(\boldsymbol{n}, \boldsymbol{\omega_i}, \boldsymbol{\omega_o}) \cdot F(\langle\, \boldsymbol{\omega_i} \,|\, \boldsymbol{\omega_h} \,\rangle)}{\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_i} \,\rangle \, \langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega_o} \,\rangle} \quad , \quad \boldsymbol{\kappa}_s \leq \mathbf{1}$$

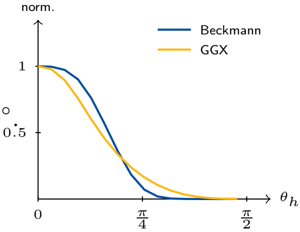


Note: In the original paper, the constant in the denominator was π and not 4. However, recent papers agree with this version.

Problem with Beckmann at grazing angles
The Beckmann NDF decays *exponentially* as $\theta_h \to 90^\circ$, producing highlights that vanish too quickly near grazing incidence. Measured BRDFs of metals, plastics, and skin show a *significantly brighter* halo in that regime.

GGX replaces the exponential with an *algebraic* $(1/x^2)$ decay, giving heavier tails that fit measurements much better:

$$D_{\text{GGX}} \propto \frac{1}{(\alpha^2 + \tan(\theta_h)^2)^2} \longrightarrow 0 \text{ slowly as } \theta_h \to 90^\circ.$$



Both at α=0.5. GGX tail persists near $\frac{\pi}{2}$.

Industry status

GGX is the **de-facto standard NDF** used in Unreal Engine, Unity HDRP, Blender Cycles, and the Academy ACES pipeline.

Definition (GGX / Trowbridge-Reitz NDF)

The isotropic *GGX Normal Distribution Function* is

$$D_{\mathsf{GGX}}(\boldsymbol{h}) = \frac{\alpha^2}{\pi \cos(\theta_h)^4 \left(\alpha^2 + \tan(\theta_h)^2\right)^2}, \qquad \alpha \in (0, 1].$$

The microfacet NDF $D(\boldsymbol{\omega}_h)$ is defined on the *hemisphere*, but **shape-invariance** is most naturally stated in *slope space* $\mathbb{R}^2$. The two are related by equating probability mass.

The slope vector. A microfacet normal $\boldsymbol{\omega}_h = (\theta_h, \phi_h)$ corresponds bijectively to the 2D slope

$$\boldsymbol{s} = \begin{pmatrix} s_x \\ s_y \end{pmatrix} = \begin{pmatrix} \tan(\theta_h)\cos(\phi_h) \\ \tan(\theta_h)\sin(\phi_h) \end{pmatrix}, \qquad |\boldsymbol{s}| = \tan(\theta_h),$$

a bijection between the upper hemisphere ($\theta_h < \pi/2$) and $\mathbb{R}^2$.

For the change of variables we need the determinant of the Jacobian J of the map $(\theta_h, \phi_h) \mapsto (s_x, s_y)$:

$$J = \begin{pmatrix} \frac{\partial s_x}{\partial \theta_h} & \frac{\partial s_x}{\partial \phi_h} \\ \frac{\partial s_y}{\partial \theta_h} & \frac{\partial s_y}{\partial \phi_h} \end{pmatrix} = \begin{pmatrix} \frac{\cos(\phi_h)}{\cos(\theta_h)^2} & -\frac{\sin(\theta_h)\sin(\phi_h)}{\cos(\theta_h)} \\ \frac{\sin(\phi_h)}{\cos(\theta_h)^2} & \frac{\sin(\theta_h)\cos(\phi_h)}{\cos(\theta_h)} \end{pmatrix}$$

From this we get $|J| = \sin(\theta_h)/\cos(\theta_h)^3$, so

$$d^2\boldsymbol{s} \;=\; \frac{\sin(\theta_h)}{\cos(\theta_h)^3}\, d\theta_h\, d\phi_h \;=\; \frac{1}{\cos(\theta_h)^3}\, d\omega_h.$$

The D–P^{22} relation: Equating the projected-area probability masses $P^{22}(\boldsymbol{s})\, d^2\boldsymbol{s} = D(\boldsymbol{\omega}_h)\cos(\theta)_h\, d\omega_h$ and substituting the Jacobian gives

$$\boxed{P^{22}(\boldsymbol{s}) \;=\; D(\boldsymbol{\omega}_h)\cos(\theta_h)^4} \qquad \Longleftrightarrow \qquad D(\boldsymbol{\omega}_h) \;=\; \frac{P^{22}(\boldsymbol{s})}{\cos(\theta_h)^4}.$$

So, the D_{GGX} corresponds to the slope distribution

$$P^{22}_{\mathsf{GGX}}(\boldsymbol{s}) = D_{\mathsf{GGX}}(\boldsymbol{\omega}_h)\cos(\theta_h)^4 = \frac{\alpha^2}{\pi\,(\alpha^2 + |\boldsymbol{s}|^2)^2}.$$

Shape-invariance – the key structural property

Definition (Shape-invariance)

An NDF D is *shape-invariant* if scaling α only rescales P^{22} without changing its shape:

$$P^{22}(\boldsymbol{s};\, \alpha) = \frac{1}{\alpha^2}\, P^{22}\Big(\frac{\boldsymbol{s}}{\alpha};\, 1\Big).$$

One verifies immediately that P^{22}_{GGX} satisfies this.

- GGX **satisfies** shape-invariance. Its slope distribution is a *Cauchy* (t_2) distribution, whose Smith $\Lambda(\boldsymbol{\omega})$ integrates to a **closed-form square root** – no special functions.
- Beckmann also satisfies shape-invariance, but its slope distribution is *Gaussian*, so $\Lambda_{\mathsf{Beckmann}}$ requires $\operatorname{erf}(\cdot)$, which has **no closed form** and must be approximated in GPU shaders via a rational-polynomial fit.
- This analytic tractability is the primary reason GGX displaced Beckmann in real-time and offline renderers.

The auxiliary function $\Lambda(\omega)$ for GGX

Recall $G_1(\boldsymbol{\omega}) = 1/(1+\Lambda(\boldsymbol{\omega}))$ where Λ measures invisible masked area per visible area.

GGX – fully analytic:

$$\boxed{\Lambda_{\mathsf{GGX}}(\boldsymbol{\omega}) = \frac{-1+\sqrt{1+\alpha^2\tan(\theta)^2}}{2}}$$

One square root, no approximation.

Therefore:

$$G_1^{\mathsf{GGX}}(\boldsymbol{\omega}) = \frac{2}{1+\sqrt{1+\alpha^2\tan(\theta)^2}}$$

Beckmann – requires erf:

$$\Lambda_{\mathsf{Beck.}}(\boldsymbol{\omega}) = \tfrac{1}{2}\left(\mathrm{erf}(a) - 1 + \frac{e^{-a^2}}{a\sqrt{\pi}}\right)$$

$$a = \frac{1}{\alpha\tan(\theta)}$$

$\Rightarrow$ rational-polynomial fit needed in shaders.

Height-correlated Smith G_2 – what shipped renderers use

Masking and shadowing are *correlated* through microfacet height: taller points are simultaneously more likely to be both visible and illuminated. This yields the *height-correlated* masking-shadowing function:

$$\boxed{G_2(\boldsymbol{\omega}_o, \boldsymbol{\omega}_i) = \frac{1}{1 + \Lambda(\boldsymbol{\omega}_o) + \Lambda(\boldsymbol{\omega}_i)}}$$

- **More accurate** than the uncorrelated product $G_1(\boldsymbol{\omega}_o) \cdot G_1(\boldsymbol{\omega}_i)$, which underestimates G_2 (most visibly when $\boldsymbol{\omega}_o = \boldsymbol{\omega}_i$).
- Still **fully analytic** for GGX: substitute Λ_{GGX} – no approximation required.
- This is the form used in **Unreal Engine 4** *(Karis 2013)*, **Frostbite** *(Lagarde & de Rousiers 2014)*, and **pbrt-v4**.

White-furnace test – linking NDF normalisation to albedo.
Consider a perfectly white microfacet surface ($\kappa_s = 1,\, F \equiv 1$) inside a uniform white environment (furnace) of unit radiance. Energy conservation requires the albedo to equal 1 for all incoming directions. **Step 1: NDF normalisation $\Rightarrow$ unit albedo for**

$G \equiv 1$. Inserting $F = 1$, $G = 1$, and the Cook-Torrance $D/4$ orientation factor into the albedo integral and cancelling yields

$$a(\boldsymbol{\omega}_i) = \int_{\Omega} \frac{D(\boldsymbol{\omega}_h)}{4\,\langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega}_i \,\rangle} \cdot \langle\, \boldsymbol{n} \,|\, \boldsymbol{\omega}_o \,\rangle \, d\omega_o = \int_{\Omega} D(\boldsymbol{\omega}_h) \cos(\theta)_h \, d\omega_h = 1,$$

where the last equality is exactly the NDF normalisation constraint. So a correctly normalised NDF guarantees energy conservation – but only if $G \equiv 1$. **Step 2: Role of**

the shadowing-masking function. Any $G < 1$ removes energy from the reflected lobe.

- The **uncorrelated** product $G_1(\boldsymbol{\omega}_o) \cdot G_1(\boldsymbol{\omega}_i)$ underestimates visibility, making rough surfaces appear too dark.
- The **height-correlated** Smith $G_2 = 1/(1 + \Lambda(\boldsymbol{\omega}_o) + \Lambda(\boldsymbol{\omega}_i))$ minimises this darkening and is the tightest upper bound consistent with a single-scattering model.
- At high roughness a residual energy deficit remains; this is addressed by multiple-scattering extensions (see next slide).

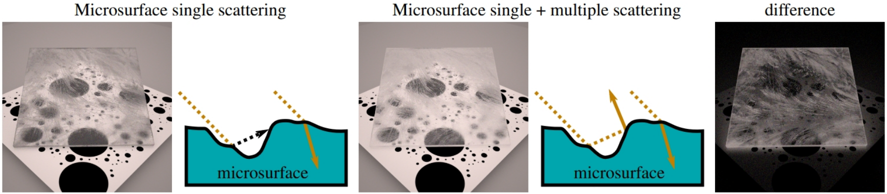


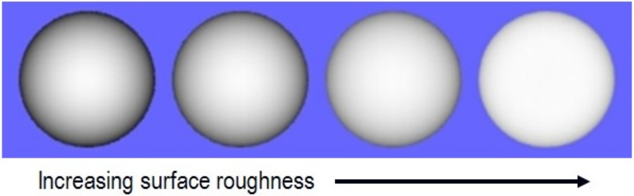
Figure: Single- (left) vs. multi-scattering (right) microfacet BRDF at high roughness.

The energy-loss problem. At high roughness many photons undergo multiple reflections between microfacets before escaping. A single-scattering model treats these photons as absorbed, so rough surfaces appear *too dark* and lose energy as $\alpha \to 1$. The white-furnace test reveals this directly: the single-scattering albedo falls below 1 at large α, even with a perfectly energy-conserving G_2. The effect is pronounced for GGX precisely because its heavier tails direct more light into high-angle inter-reflections.

Multiple-scattering BRDF. **heitz2016multiple** derive a physically correct multiple-scattering BSDF for any shape-invariant NDF under the Smith model – including both **GGX** and Beckmann. The method simulates a random walk on the microsurface. The additional lobe is diffuse-like, broadens with roughness, and restores energy conservation without any ad-hoc correction.

Observation: The moon reflects diffusely but does not have dark borders.

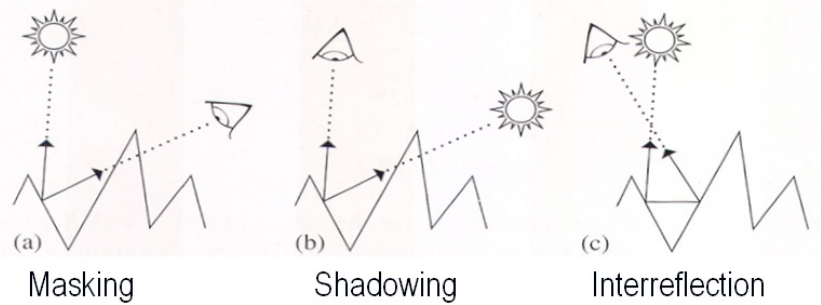
Only "geometrically smooth" (wavelength of the light) matte surfaces reflect ideal diffuse (lambertian)!



Idea: Create a microfacet model similar to the Cook-Torrance model to describe rough diffuse surfaces.

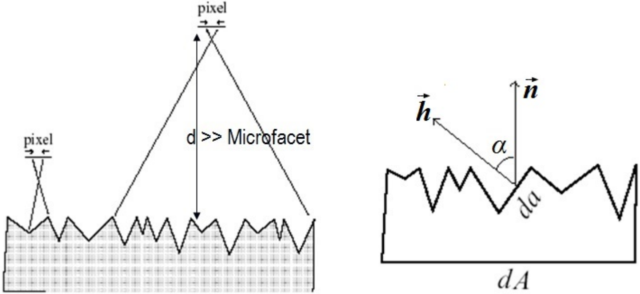
$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \frac{\boldsymbol{\kappa}_s}{4} \frac{D(\boldsymbol{h}) \cdot G(\boldsymbol{n}, \boldsymbol{l}, \boldsymbol{v}) \cdot F(\langle \boldsymbol{l} \,|\, \boldsymbol{h} \rangle)}{\langle \boldsymbol{n} \,|\, \boldsymbol{l} \rangle \langle \boldsymbol{n} \,|\, \boldsymbol{v} \rangle}$$

So the same geometrically effects should be considered:



Note: Here, inter-reflections are also considered (but only first order).

But the microfacets reflect diffusely and not according to Snell's law!



Finally, the Normal Distribution Function is chosen to be a Gaussian distribution

$$D(\boldsymbol{h}) = \mathcal{N}(\alpha \mid 0\,, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{\alpha^2}{2\sigma^2}}$$

with angle $\alpha = \cos(\langle\, \boldsymbol{n} \,|\, \boldsymbol{h} \,\rangle)^{-1}$ and roughness parameter σ. This leads to the BRDF for rough diffuse surfaces.

This leads to the BRDF for rough diffuse surfaces.

Definition (Oren-Nayar-BRDF[17])

The *Oren-Nayar-BRDF* is defined as

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \frac{\boldsymbol{\kappa}_d}{\pi}(A + B \max\{0\,,\, \cos(\phi_i - \phi_o)\} \sin(\alpha) \tan(\beta)) \quad , \quad \boldsymbol{\kappa}_d \leq \mathbf{1}$$

where

$$A = 1 - \frac{\sigma^2}{2\,(\sigma^2 + 0.33)} \quad , \quad B = \frac{0.45\,\sigma^2}{\sigma^2 + 0.09}$$
$$\alpha = \max\{\theta_i\,,\, \theta_o\} \quad , \quad \beta = \min\{\theta_i\,,\, \theta_o\}$$

Note: Using a variance of $\sigma^2 = 0$, which means a perfectly smooth surface, we get the values

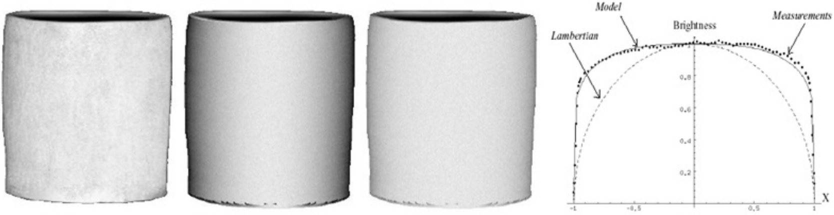
$$A = 1 \quad , \quad B = 0$$

and the Oren-Nayar-BRDF reduces to the Lambert-BRDF!

[17] M. Oren and S. K. Nayar. "Generalization of Lambert's reflectance model". In: *Proceedings of the 21st annual conference on Computer graphics and interactive techniques*. ACM. 1994, pp. 239–246.

Oren-Nayar-BRDF

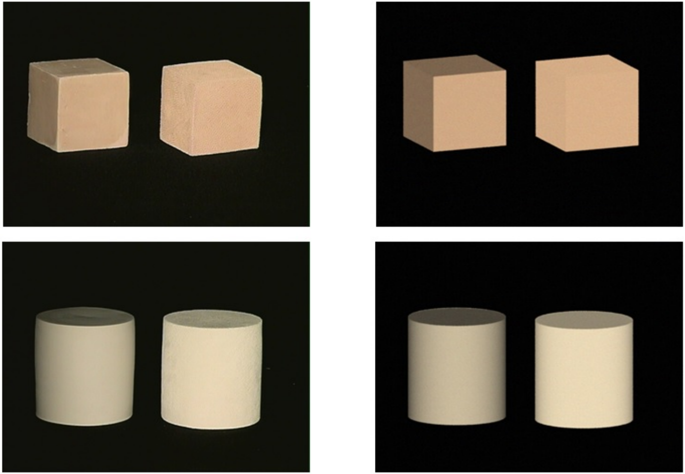
Results:



(a) Image (b) Lambertian (c) Model (d)

Fig. 7. (a-c) Real image of a cylindrical clay vase compared with images rendered using the Lambertian and proposed models. Illumination is from the direction $\theta_i = 0°$. (d) Comparison between image brightness along the cross-sections of the three vases.

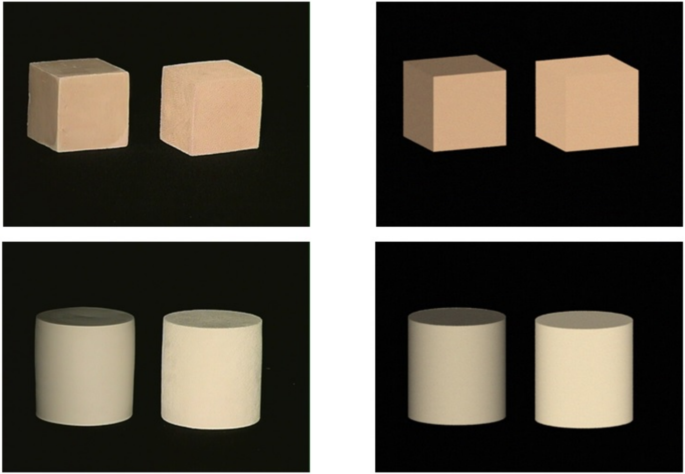
Results:



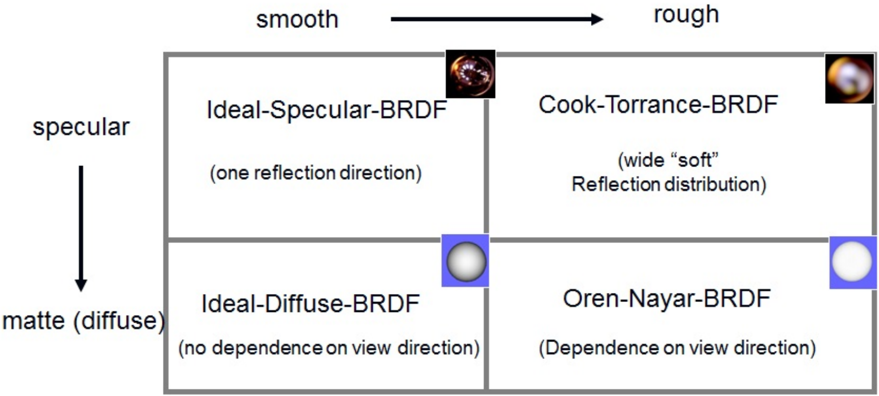
Real Images Renderings

Oren-Nayar-BRDF

Results:



Real Images Renderings

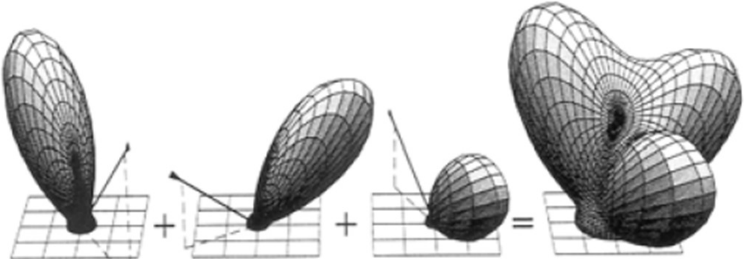

smooth
rough
specular
matte (diffuse)
Ideal-Specular-BRDF
(one reflection direction)
Cook-Torrance-BRDF
(wide “soft”
Reflection distribution)
Ideal-Diffuse-BRDF
(no dependence on view direction)
Oren-Nayar-BRDF
(Dependence on view direction)

Model	**Recip.**	**Phys.**	**Aniso.**	**Dim.**	**Key limitation**
Lambert	✓	✓	–	0	Too flat for real surfaces
Oren–Nayar	✓	✓	–	1	Diffuse only
Phong (original)	–	–	–	1	Not reciprocal
Cosine-Lobe	✓	✓	–	1	Lobe centred at $\boldsymbol{r}$ only
Lafortune	✓	(✓)	✓	varies	Requires nonlinear fitting
Gauss-BRDF	✓	–	–	1	Energy not conserved
Cook–Torrance	✓	✓	✓	2	Single scattering only
Cook–Torrance/GGX	✓	✓	✓	2	Single scattering only
Data-driven	✓	✓	✓	3–4	Storage; interpolation

Columns: Reciprocal · Physically plausible · Supports anisotropy · Number of free roughness parameters.

Note: Data-driven models (MERL database, Rusinkiewicz parameterisation) are covered in the next lecture.

Many (analytical) BRDF models are composed of several individual components.



Therefore, diffuse models ($\boldsymbol{\kappa}_d$) and specular models ($\boldsymbol{\kappa}_s$) are combined using superposition to get a full reflection model.

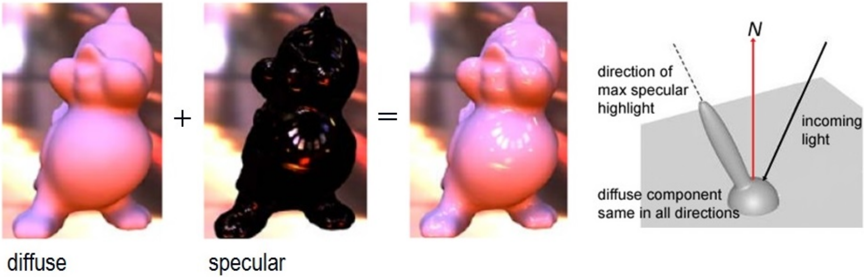
Example: Naturally, materials that can be described by the Cosine-Lobe-BRDF also reflect diffusely which leads to the BRDF

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l}, \boldsymbol{v}) = \frac{\boldsymbol{\kappa}_d}{\pi} + \boldsymbol{\kappa}_s \cdot \frac{m+2}{2\pi} \langle \boldsymbol{r} \,|\, \boldsymbol{v} \rangle^m \quad , \quad \boldsymbol{\kappa}_d + \boldsymbol{\kappa}_s \leq \mathbf{1}$$

Composite BRDF's

Example: Naturally, materials that can be described by the Cosine-Lobe-BRDF also reflect diffusely which leads to the BRDF

$$\boldsymbol{f}_{BRDF}(\boldsymbol{l},\boldsymbol{v}) = \frac{\boldsymbol{\kappa}_d}{\pi} + \boldsymbol{\kappa}_s \cdot \frac{m+2}{2\pi} \langle\, \boldsymbol{r} \,|\, \boldsymbol{v} \,\rangle^m \quad , \quad \boldsymbol{\kappa}_d + \boldsymbol{\kappa}_s \leq \mathbf{1}$$



Note: Some BRDF's are defined directly with diffuse and specular components (Cook-Torrance , . . .). Here, we defined them only using its most important component!

[1] M. Ashikmin, S. Premože, and P. Shirley. "A microfacet-based BRDF generator". In: *Proceedings of the 27th annual conference on Computer graphics and interactive techniques*. ACM Press/Addison-Wesley Publishing Co. 2000, pp. 65–74.

[2] P. Beckmann and A. Spizzichino. "The scattering of electromagnetic waves from rough surface". In: (1963).

[3] J. F. Blinn. "Models of light reflection for computer synthesized pictures". In: *ACM SIGGRAPH Computer Graphics*. Vol. 11. 2. ACM. 1977, pp. 192–198.

[4] R. L. Cook. "A Reflectance Model for Computer Graphics". en. In: *ACM Transactions on Graphics* 1.1 (), p. 18.

[5] R. L. Cook and K. E. Torrance. "A reflectance model for computer graphics". In: *ACM Transactions on Graphics (TOG)* 1.1 (1982), pp. 7–24.

[6] E. Heitz. "Generating procedural beckmann surfaces". In: *Tech. rep., Unity Technologies* (2015).

[7] E. Heitz. "Understanding the masking-shadowing function in microfacet-based BRDFs". In: (2014).

[8] E. P. Lafortune, S.-C. Foo, K. E. Torrance, and D. P. Greenberg. “Non-linear approximation of reflectance functions”. In: *Proceedings of the 24th annual conference on Computer graphics and interactive techniques*. ACM Press/Addison-Wesley Publishing Co. 1997, pp. 117–126.

[9] J. Lambert and E. Anding. *Photometria, sive De mensura et gradibus luminis, colorum et umbrae (1760)*. Ostwalds Klassiker der exakten Wissenschaften. W. Engelmann, 1892.

[10] R. R. Lewis. “Making shaders more physically plausible”. In: *Computer Graphics Forum*. Vol. 13. 2. 1994, pp. 109–120.

[11] M. Minnaert. “The reciprocity principle in lunar photometry”. In: *The Astrophysical Journal* 93 (1941), pp. 403–410.

[12] M. Oren and S. K. Nayar. “Generalization of Lambert’s reflectance model”. In: *Proceedings of the 21st annual conference on Computer graphics and interactive techniques*. ACM. 1994, pp. 239–246.

[13] B. T. Phong. “Illumination for computer generated pictures”. In: *Communications of the ACM* 18.6 (1975), pp. 311–317.

[14] M. Ribardière, B. Bringier, L. Simonot, and D. Meneveaux. "Microfacet BSDFs generated from NDFs and explicit microgeometry". In: *ACM Transactions on Graphics (TOG)* 38.5 (2019), pp. 1–15.

[15] P. S. Shirley. "PHYSICALLY BASED LIGHTING CALCULATIONS FOR COMPUTER GRAPHICS". en. In: (), p. 183.

[16] B. Smith. "Geometrical shadowing of a random rough surface". In: *IEEE transactions on antennas and propagation* 15.5 (1967), pp. 668–671.

[17] K. E. Torrance and E. M. Sparrow. "Theory for off-specular reflection from roughened surfaces". In: *JOSA* 57.9 (1967), pp. 1105–1112.

[18] T. Trowbridge and K. P. Reitz. "Average irregularity representation of a rough surface for ray reflection". In: *JOSA* 65.5 (1975), pp. 531–536.

[19] B. Walter, S. R. Marschner, H. Li, and K. E. Torrance. "Microfacet Models for Refraction through Rough Surfaces.". In: *Rendering techniques* 2007 (2007), 18th.